Team RoboManipal

Objective

We are building NurseryHub, an offline edge AI system that monitors every plant as an individual. We will design and implement a system capable of monitoring, maintaining, and controlling environmental conditions for plants within a nursery.

Abstract and Plan

India has over 50,000 registered nurseries, the majority operating with fewer than five staff. Ornamental horticulture, tissue culture labs, bonsai collections, and grafting research stations all face the same constraint: biological systems with individual needs, managed by people whose attention is finite. GroveOS is not a gadget for a single plant on a windowsill. It is infrastructure for anyone who grows living things at scale and cannot afford to lose them.

A Maidenhair fern (Adiantum) can collapse in four hours when its soil dries below 60% in a nursery, whereas the Haworthia two benches over can tolerate 15% moisture for two weeks without complaining. The rootstock of a grafted rose requires a different level of hydration than the scion. If the temperature rises by two degrees above its comfortable range, a bud-set gardenia will abort every flower.

Soil moisture, temperature, humidity, ethylene, and CO2 are continuously measured by a network of modularized MCU nodes. The SBC hub fits a moisture decay curve to each plant's history to predict "water in 4 hours" rather than "water now," applies per-species thresholds (Calathea at 18% moisture fires an alert while cactus at 18% does not), and spreads disease risk through a spatial adjacency graph so that neighbors of a sick plant are flagged proactively before symptoms appear.

Sensors Sub Category Used-

- Temperature & Humidity
- Light & Colour
- Sound & Vibration
- Gas & Air Quality
- Soil & Moisture

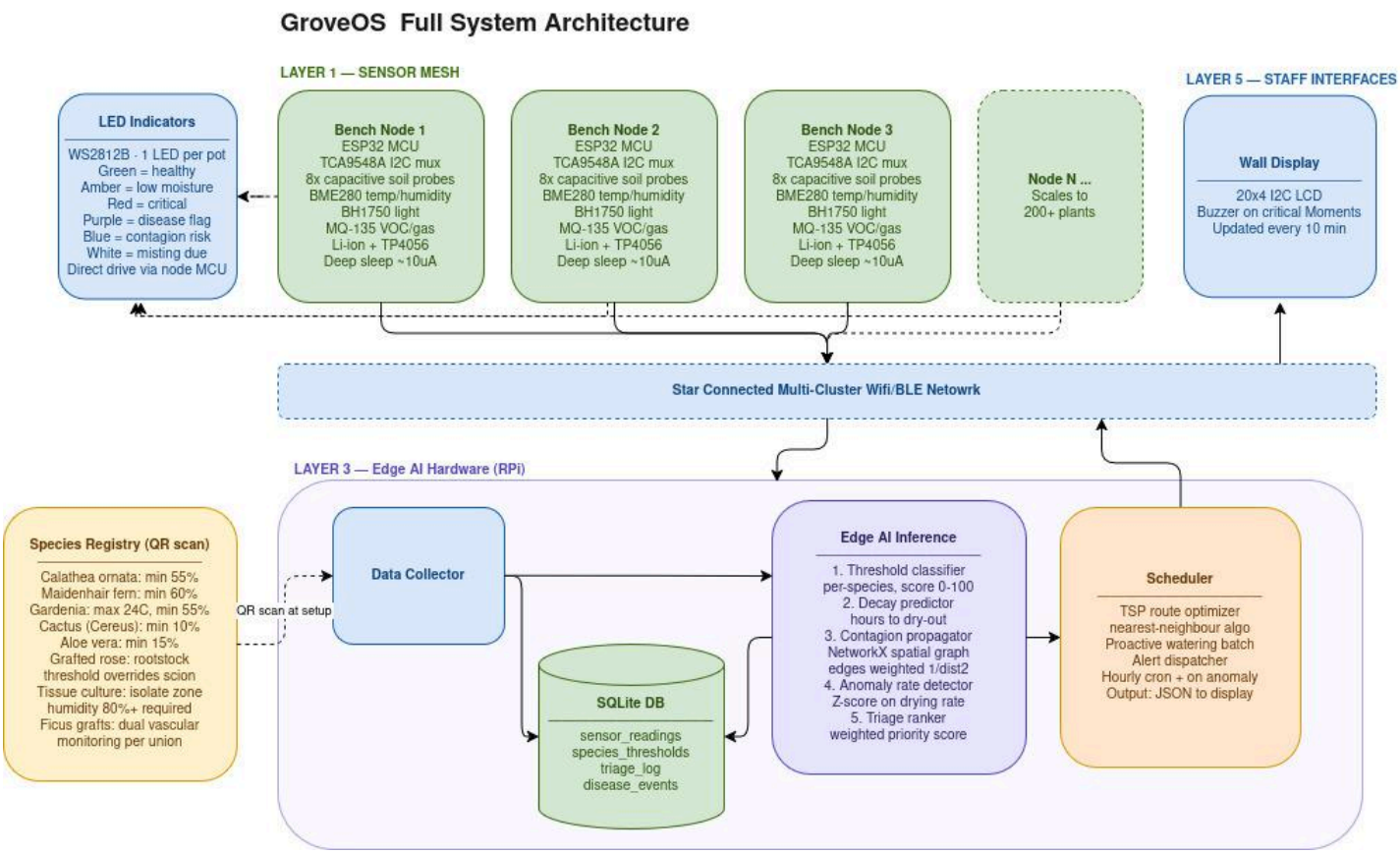
Every morning the hub outputs a numbered walk route, the shortest path through all flagged plants, so two staff manage 200 plants purposefully instead of wandering.

The expected outcome is a nursery operating on a predictive rather than reactive model, with alert fatigue eliminated, disease spread caught early, and daily inspection time cut from two hours to under thirty minutes.

Concept Category

- Microcontroller - ESP32
- SBC - Raspberry Pi
- Sensors - soil/moisture, temperature/humidity, gas/air quality
- Communication - Wi-Fi and BT
- Display - LCD
- Output Devices - addressable LED strips.

Block Diagram



Adaptability Statement

There are multiple parameters to judge plant health hence the sensors can be changed depending upon the need. If soil probes are unavailable we shift to temperature and gas sensors as primary health proxies.

If the product category shifts away from agriculture entirely, the same architecture distributed sensor nodes, a central inference hub, a route optimised output applies directly to warehouse inventory monitoring, server room environmental management, or hospital bed condition tracking.